

**INSTITUTE OF ENGINEERING AND TECHNOLOGY
DEEN DAYAL UPADHYAYA GORAKHPUR UNIVERSITY
GORAKHPUR**



**COURSE STRUCTURE
&
SYLLABUS**

FOR

B.TECH.

in

Artificial Intelligence and Machine Learning

Based

on

AICTE MODEL CURRICULUM

[Effective from the Session: 2025-26]

Semester-IV (Second Year) Curriculum

S. No.	Type of Course	Course Code	Course Title	Credit
1.	Professional Core Course	AIML206	Soft Computing	3+0
2.	Professional Core Course	AIML207	Operating Systems	4+0
3.	Professional Core Course	AIML208	AI Ethics and Society	2+0
4.	Professional Core Course	AIML209	Machine Learning	4+0
5.	Professional Core Course	AIML259	Machine Learning Lab	0+1
6.	Professional Core Course	AIML210	Theory of Automata and Formal Languages	4+0
7.	Professional Core Course	AIML256	Soft Computing Lab	0+1
8.	Professional Core Course	AIML257	Operating System Lab	0+1
9.	AEC Course	AEC-x	AEC Course	02
Total Credit				22

Soft Computing (AIML206)

Credit: 3+0

Course Objective: This course enables students to adopt nature-inspired computing models to solve specific problems.

Syllabus

1. Introduction: What is computational intelligence?- Biological basis for neural networks- Biological versus Artificial neural networks- Biological basis for evolutionary computation- Behavioral motivations for fuzzy logic, Myths about computational intelligence- Computational intelligence application areas, Evolutionary computation, computational intelligence-Adoption, Types, self-organization and evolution, Historical views of computational intelligence, Computational intelligence and Soft computing versus Artificial intelligence and Hard computing.
2. Genetic Algorithm, Genetic Representations, Initial Population, Fitness Function, Selection and Reproduction, Genetic Operators(Selection, Crossover, Mutation), Artificial Immune Systems, Other Algorithms Harmony Search, Honey-Bee Optimization, Memetic Algorithms, Co-evolution, Multi Objective Optimization, Artificial Life, Constraint Handling, Collective Systems Collective Behaviour and Swarm Intelligence, Particle Swarm Optimization and Ant Colony Optimization, Artificial evolution of Competing Systems, Artificial Evolution of cooperation and competition. Recent topics from research papers.
3. Fuzzy Logic-I (Introduction): Basic concepts of fuzzy logic, Fuzzy sets and Crisp sets, Fuzzy set theory and operations, Properties of fuzzy sets, Fuzzy and Crisp relations, Fuzzy to Crisp conversion. Fuzzy Logic –II (Fuzzy Membership, Rules): Membership functions, interference in fuzzy logic, fuzzy if-then rules, Fuzzy implications and Fuzzy algorithms, Fuzzy fications & Defuzzificataions, Fuzzy Controller, Industrial applications
4. Neural Network Concepts and Paradigms- What Neural Networks are? Why they are useful, Neural network components and terminology- Topologies - Adaptation, Comparing neural networks and other information Processing methods- Stochastic- Kalman filters , Perceptron Network, Adaline Network, Madaline Network, Back Propagation Network, Linear and Nonlinear regression - Correlation - Bayes classification -Vector quantization - Radial basis functions Preprocessing - Post processing.

References

1. Dario Floreano, Claudio Mattiussi, "Bio-Inspired Artificial Intelligence: Theories, Methods and Technologies", MIT Press, 2008.
2. Eberhart, E. and Y. Shi., "Coputational Intelligence: Concepts and Implementations", Morgan Kauffmann, San Diego, 2007.
3. D. E. Goldberg, "Genetic algorithms in search, optimization, and machine learning", AddisonWesley, 1989
4. R. C. Ebelhart et al., "Swarm Intelligence", Morgan Kaufmann, 2001. • M. Dorigo and T. Stutzle, "Ant Colony Optimization", A Bradford Book, 2004.
5. Leandro Nunes De Castro, Fernando Jose Von Zuben, "Recent Developments in Biologically Inspired Computing", Idea Group Publishing, 2005.

Soft Computing Lab (AIML256)

Credit: 0+1

1. Implement Perceptron Network
2. Implement Adaline Network
3. Implement Madaline Network for XOR Function
4. Implement Back Propagation Network for XOR Function using Bipolar Inputs and Binary Targets.
5. Implement Kohonen Self-Organizing Feature Map
6. Genetic Algorithm Implementation to Maximize $F(X_1, X_2) = 4X_1 + 3X_2$
7. Genetic Algorithm Implementation to maximize $F(X) = X^2$
8. Genetic Algorithm Implementation to minimize $F(X) = X^2$
9. Implementation of Traveling Salesman Problem.
10. Genetic Algorithm Implementation to Minimize $F(X_1, X_2) = 4X_1 + 3X_2$

Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

Operating Systems (AIML207)

Credit: 4+0

Course Objective: Student will learn about processes and processor management, concurrency and synchronization, memory management schemes, and secondary storage management, security and protection.

Syllabus

- 1. Introduction:** Operating system and functions, Classification of Operating systems- Batch, Interactive, Time-sharing, Real-Time System, Multiprocessor Systems, Multiuser Systems, Multiprocess Systems, Multithreaded Systems, Operating System Structure- Layered structure, System Components, Operating System services, Reentrant Kernels, Monolithic and Microkernel Systems.
- 2. Concurrent Processes:** Process Concept, Principle of Concurrency, Producer / Consumer Problem, Mutual Exclusion, Critical Section Problem, Dekker's solution, Peterson's solution, Semaphores, Test and Set operation; Classical Problem in Concurrency- Dining Philosopher Problem, Sleeping Barber Problem; Inter Process Communication models and Schemes, Process generation.
- 3. CPU Scheduling & Disk Scheduling:** Scheduling Concepts, Performance Criteria, Process States, Process Transition Diagram, Schedulers, Process Control Block (PCB), Process address space, Process identification information, Threads and their management, Scheduling Algorithms, Multiprocessor Scheduling. Deadlock: System model, Deadlock characterization, Prevention, Avoidance and detection, Recovery from deadlock. Disk Scheduling: I/O devices, and I/O subsystems, I/O buffering, Disk storage and disk scheduling.
- 4. Memory Management:** Basic bare machine, Resident monitor, Multiprogramming with fixed partitions, Multiprogramming with variable partitions, Protection schemes, Paging, Segmentation, Paged segmentation, Virtual memory concepts, Demand paging, Performance of demand paging, Page replacement algorithms, Thrashing, Cache memory organization, Locality of reference.

Course Outcome (CO): At the end of course, the student will be able to:

CO1: Understand the structure and functions of OS.

CO2: Learn about Processes, Threads and Scheduling algorithms.

CO3: Understand the principles of concurrency and Deadlocks.

CO4: Learn various memory management schemes.

References

1. Silberschatz, Galvin and Gagne, "Operating Systems Concepts", Wiley.
2. Sibsanakar Halder and Alex A Aravind, "Operating Systems", Pearson Education.
3. Harvey M Dietel, "An Introduction to Operating System", Pearson Education.
4. D M Dhamdhare, "Operating Systems: A Concept based Approach", 2nd Edition, TMH.
5. William Stallings, "Operating Systems: Internals and Design Principles", 6th Edition, Pearson Education.

Operating System Lab (AIML257)

Credit: 0+1

1. Study of hardware and software requirements of different operating systems (UNIX, LINUX, WINDOWS XP/ WINDOWS 7/8/10)
2. Execute various UNIX system calls for
 - a. Process management
 - b. File management
 - c. Input/output Systems calls
3. Implement CPU Scheduling Policies:
 - a. SJF
 - b. Priority
 - c. FCFS
 - d. Multi-level Queue
4. Implement file storage allocation technique:
 - a. Contiguous (using array)
 - b. Linked -list (using linked-list)
 - c. Indirect allocation (indexing)
5. Implementation of contiguous allocation techniques:
 - a. Worst-Fit
 - b. Best- Fit
 - c. First- Fit
6. Calculation of external and internal fragmentation
 - a. Free space list of blocks from system
 - b. List process file from the system
7. Implementation of compaction for the continually changing memory layout and calculate total movement of data.
8. Implementation of resource allocation graph RAG).
9. Implementation of Banker's algorithm.
10. Conversion of resource allocation graph (RAG) to wait for graph (WFG) for each type of method used for storing graph.
11. Implement the solution for Bounded Buffer (producer-consumer) problem using inter process communication techniques-Semaphores.
12. Implement the solutions for Readers-Writer's problem using inter process communication technique -Semaphore.

Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

AI Ethics and Society (AIML208)

Credit: 2+0

Course Objective:

To introduce students to the ethical principles and societal impact of AI technologies. To analyze bias, fairness, and privacy issues in AI models and data-driven decision-making. To explore AI's role in society, including its impact on employment, governance, and cultural aspects.

Syllabus

1. Foundations of AI Ethics

Introduction to AI and Ethics: Definition and scope of AI ethics, historical context of AI development, importance of ethical considerations in AI, ethical challenges in machine learning and automation, AI's impact on decision-making and human autonomy. Ethical Theories and AI: Utilitarianism, deontology, virtue ethics, fairness, accountability, and transparency (FAT) in AI, principles of responsible AI development, ethical dilemmas in AI applications, trade-offs between privacy and utility.

2. Bias, Fairness, and Privacy in AI

Bias and Fairness in AI: Sources of bias in AI models, algorithmic fairness and discrimination, mitigating bias in data and model training, fairness-aware machine learning techniques, real-world examples of biased AI decisions. Privacy and Surveillance: Data privacy in AI systems, ethical concerns in data collection and processing, GDPR and AI regulations, AI-driven surveillance and its societal implications, privacy-preserving AI techniques (differential privacy, federated learning).

3. Societal Impact and AI Governance

AI and Society: AI's impact on employment and the future of work, AI in healthcare, finance, and education, social and cultural implications of AI-driven automation, ethical concerns in AI-generated content and misinformation, AI and sustainability. AI Governance and Regulation: Global AI policies and frameworks, role of government and industry in AI regulation, AI ethics guidelines by organizations (EU, IEEE, UNESCO), case studies of AI governance failures, ethical AI deployment in businesses and public services.

4. Ethical AI Design and the Future of AI

Ethical AI Development and Human-Centered AI: Explainable AI (XAI) and interpretability, human-in-the-loop AI systems, designing AI for social good, ethical AI design principles, AI for accessibility and inclusion. The Future of AI and Ethical Challenges: Emerging ethical challenges in AI (AGI, autonomous weapons, deepfake technology), ethical concerns in AI-driven decision-making, philosophical perspectives on AI consciousness, ensuring ethical AI development for future generations.

Course Outcome (CO): Upon completion of the course, the students should be able to

CO1: To introduce students to the ethical principles and societal impact of AI technologies.

CO2: To analyse bias, fairness, and privacy issues in AI models and data-driven decision-making.

CO3: To explore AI's role in society, including its impact on employment, governance, and cultural aspects.

CO4: To understand AI policies, regulations, and global ethical frameworks for responsible AI development.

References

1. Mark Coeckelbergh – *AI Ethics*, MIT Press, 2020.
2. Nick Bostrom – *Superintelligence: Paths, Dangers, Strategies*, Oxford University Press, 2014.
3. Virginia Eubanks – *Automating Inequality: How High-Tech Tools Profile, Police, and Punish the Poor*, St. Martin's Press, 2018.
4. Kate Crawford – *Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*, Yale University.

Machine Learning (AIML209)

Credit: 4+0

Course Objectives: This course provides an advanced level of understanding to machine learning and statistical pattern recognition. It offers some of the most cost-effective approaches to automated knowledge acquisition in emerging data-rich disciplines and focuses on the theoretical understanding of these methods, as well as their computational implications

Syllabus

- 1. Introduction:** Learning, Types of Learning, well defined learning problems, Designing a Learning System, History of ML, Introduction of Machine Learning Approaches – (Artificial Neural Network, Clustering, Reinforcement Learning, Decision Tree Learning, Bayesian networks, Support Vector Machine, Genetic Algorithm), Issues in Machine Learning and Data Science Vs Machine Learning.
- 2. Regression and Decision Tree Learning:** Linear Regression and Logistic Regression
BAYESIAN LEARNING - Bayes theorem, Concept learning, Bayes Optimal Classifier, Naïve Bayes classifier, Bayesian belief networks, EM algorithm. SUPPORT VECTOR MACHINE: Introduction, Types of support vector kernel – (Linear kernel, polynomial kernel, and Gaussian kernel), Hyperplane – (Decision surface), Properties of SVM, and Issues in SVM. DECISION TREE LEARNING: Decision tree learning algorithm, Inductive bias, Inductive inference with decision trees, Entropy and information theory, Information gain, ID-3 Algorithm, Issues in Decision tree learning. INSTANCE-BASED LEARNING – k-Nearest Neighbor Learning, Locally Weighted Regression, Radial basis function networks, Case-based learning.
- 3. Artificial Neural Networks:** Perceptron's, Multilayer perceptron, Gradient descent and the Delta rule, Multilayer networks, Derivation of Backpropagation Algorithm, Generalization, Unsupervised Learning – SOM Algorithm and its variant; DEEP LEARNING - Introduction, concept of convolutional neural network , Types of layers – (Convolutional Layers , Activation function , pooling , fully connected) , Concept of Convolution (1D and 2D) layers, Training of network, Case study of CNN for e.g. on Diabetic Retinopathy, Building a smart speaker, Self-driving car etc.
- 4. Reinforcement Learning:** Introduction to Reinforcement Learning, Learning Task, Example of Reinforcement Learning in Practice, Learning Models for Reinforcement – (Markov Decision process, Q Learning - Q Learning function, Q Learning Algorithm), Application of Reinforcement Learning, Introduction to Deep Q Learning. GENETIC ALGORITHMS: Introduction, Components, GA cycle of reproduction, Crossover, Mutation, Genetic Programming, Models of Evolution and Learning, Applications.

Course Outcome (CO): At the end of course, the student will be able to

CO1: Understand the need for machine learning for various problem solving.

CO2: Understand a wide variety of learning algorithms and how to evaluate models generated from data.

CO3: Understand the latest trends in machine learning.

CO4: Design appropriate machine learning algorithms and apply the algorithms to a real-world problem.

References

1. Tom M. Mitchell, —Machine Learning, McGraw-Hill Education (India) Private Limited, 2013.
2. Ethem Alpaydin, —Introduction to Machine Learning (Adaptive Computation and Machine Learning), The MIT Press 2004.
3. Stephen Marsland, —Machine Learning: An Algorithmic Perspective, CRC Press, 2009.
4. Bishop, C., Pattern Recognition and Machine Learning. Berlin: Springer-Verlag.

Machine Learning Lab (AIML259)

Credit: 0+1

1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file.
2. For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm to output a description of the set of all hypotheses consistent with the training examples.
3. Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.
4. Build an Artificial Neural Network by implementing the Back propagation algorithm and test the same using appropriate data sets.
5. Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.
6. Assuming a set of documents that need to be classified, use the naïve Bayesian Classifier model to perform this task. Built-in python classes/API can be used to write the program. Calculate the accuracy, precision, and recall for your data set.
7. Write a program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using standard Heart Disease Data Set. You can use Python ML library classes/API.
8. Apply EM algorithm to cluster a set of data stored in a .CSV file. Use the same data set for clustering using k-Means algorithm. Compare the results of these two algorithms and comment on the quality of clustering. You can add Python ML library classes/API in the program.
9. Write a program to implement k-Nearest Neighbor algorithm to classify the iris data set. Print both correct and wrong predictions. Python ML library classes can be used for this problem.
10. Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select appropriate data set for your experiment and draw graphs.
11. Carry out the performance analysis of classification algorithms on a specific dataset.

Note: The instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

Theory of Automata and Formal Languages (AIML210)

Credit: 4+0

Course Objective: This course focuses on the basic theory of Computer Science and formal methods of computation like automata theory, formal languages, grammars and Turing Machines. The objective of this course is to explore the theoretical foundations of computer science from the perspective of formal languages and classify machines by their power to recognize languages.

Syllabus

- 1. Basic Concepts and Automata Theory:** Introduction to Theory of Computation- Automata, Computability and Complexity, Alphabet, Symbol, String, Formal Languages, Deterministic Finite Automaton (DFA)- Definition, Representation, Acceptability of a String and Language, Non Deterministic Finite Automaton (NFA), Equivalence of DFA and NFA, NFA with ϵ -Transition, Equivalence of NFA's with and without ϵ -Transition, Finite Automata with output- Moore Machine, Mealy Machine, Equivalence of Moore and Mealy Machine, Minimization of Finite Automata, Myhill-Nerode Theorem, Simulation of DFA and NFA.
- 2. Regular Expressions and Languages:** Regular Expressions, Transition Graph, Kleene's Theorem, Finite Automata and Regular Expression- Arden's theorem, Algebraic Method Using Arden's Theorem, Regular and Non-Regular Languages- Closure properties of Regular Languages, Pigeonhole Principle, Pumping Lemma, Application of Pumping Lemma, Decidability- Decision properties, Finite Automata and Regular Languages, Regular Languages and Computers, Simulation of Transition Graph and Regular language.
- 3. Regular and Non-Regular Grammars:** Context Free Grammar (CFG)-Definition, Derivations, Languages, Derivation Trees and Ambiguity, Regular Grammars-Right Linear and Left Linear grammars, Conversion of FA into CFG and Regular grammar into FA, Simplification of CFG, Normal Forms- Chomsky Normal Form (CNF), Greibach Normal Form (GNF), Chomsky Hierarchy, Programming problems based on the properties of CFGs.
- 4. Push Down Automata and Properties of Context Free Languages:** Nondeterministic Pushdown Automata (NPDA)- Definition, Moves, A Language Accepted by NPDA, Deterministic Pushdown Automata (DPDA) and Deterministic Context free Languages (DCFL), Pushdown Automata for Context Free Languages, Context Free grammars for Pushdown Automata, two stack Pushdown Automata, Pumping Lemma for CFL, Closure properties of CFL, Decision Problems of CFL, Programming problems based on the properties of CFLs. Turing Machines and Recursive Function Theory: Basic Turing Machine Model, Representation of Turing Machines, Language Acceptability of Turing Machines, Techniques for Turing Machine Construction, Modifications of Turing Machine, Turing Machine as Computer of Integer Functions, Universal Turing machine, Linear Bounded Automata, Church's Thesis, Recursive and Recursively Enumerable language, Halting Problem, Post's Correspondence Problem, Introduction to Recursive Function Theory.

Course Outcome (CO): At the end of course, the student will be able to

CO1: Analyse and design finite automata, pushdown automata, Turing machines, formal languages.

CO2: Analyse and design, Turing machines, formal languages, and grammars.

CO3: Demonstrate the understanding of key notions, such as algorithm, computability, decidability, and complexity through problem solving.

CO4: Prove the basic results of the Theory of Computation.

References

1. Introduction to Automata theory, Languages and Computation, J.E.Hopcraft, R.Motwani, and Ullman. 2nd edition, Pearson Education Asia
2. Introduction to languages and the theory of computation, J Martin, 3rd Edition, Tata McGraw Hill
3. Elements and Theory of Computation, C Papadimitrou and C. L. Lewis, PHI
4. Mathematical Foundation of Computer Science, Y.N.Singh, New Age International.